

Lab 1

	Bubble Sort	Quick Sort
Execution Time	0.954s	0.004s
CPU Cycles	3888290062	Negligible
Instructions	5855016645	Negligible
IPC	1.51	Similar CPU efficiency
Cache References	87636	Negligible
Cache Misses	14192	-
Hotspot Function	Bubble Sort function 100%	-

Questions

1) Which sorting algorithm performed better? Explain your observation.

Ans: Quick sort performed better than bubble sort. For dataset with 18250 values bubble sort took 0.954s, quick sort was able to do 238times faster.

2) Which function consumed the maximum execution time?

Ans: Bubble sort took more time according to GPROF report. It performs nested loops and repeated comparisons with adjacent elements resulting in large number of operations. So, bubble sort is the bottleneck.

3) How does execution time change with increasing input?

Ans: As the input size increases, the execution time changes for both the algorithms at different rates. For bubble sort it increases drastically as it is $O(N^2)$ time complexity, for quick sort it increases more slowly as average complexity is $O(N\log N)$.

4) How did compiler optimizations (-O2 and -O3) improve performance?

Ans: Compiler optimizations (-O2 and -O3) generates efficient code without changing the program logic. It optimizes loop, removes dead codes, reorders instructions. -O3 applies more aggressive optimizations on -O2, so drastically reducing the execution time, reducing CPU cycles and increasing efficiency of CPU.

5) Which algorithm would you recommend for large datasets? Justify your answer.

Ans: I would recommend Quick sort for large datasets. It is faster in execution compared to Bubble sort, GPROF report shows that bubble sort is the primary bottleneck, whereas quick sort completes in negligible time.